MIL-E-1/1358A
27 April 1972
SUPERSEDING
MIL-E-1/1358 (Sig C)
28 July 1960

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 3B4WA

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: RF, beam power amplifier, F1 = 100 MHz

Outline	---	5-2 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated filament (See note 1)

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g2	fct, g3, int sd (see note 2)	g1	f-	f+	fct, g3, int sd (see note 2)	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ic1	Pa	Pg2	TE	(A) Alt
Unit:	V	Vdc	Vdc	Vdc	mAdc	mAdc	W	W	°C	ft
Maximum:	1.44 or 2.88 (prl) (ser)	150	---	135	25	1.5	3.0	1.1	180	Note 3
Minimum:	1.06 or 2.12 (prl) (ser)	---	-75	---	---	---	---	---	---	---
TEST CONDITIONS:	2.5 Vac	200	-25	150	---	---	---	---	---	---

GENERAL

Qualification - Required

(A) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
1202	Filament burnout	-		---	---	---	---	---	---
	Quality conformance inspection, part 1								
1266	Total grid current	4		0.65	II	Ic1	0	-1.5	µAdc
1256	Electrode current (anode)	-		0.65	II	Ib	13	26	mAdc
1256	Electrode current (screen)	-		0.65	II	Ic2	---	2	mAdc
---	Operation peak output voltage	Fig 1	Ebb = 150 Vdc; Ec1 = 0; Ec2 = 135 Vdc; Rl = 1,000 ohms; Eg1 = 50 Vac; Rg1 = 55,000 ohms	0.65	II	ep	100	---	V
---	Operation screen grid current	Fig 1	Ebb = 150 Vdc; Ec1 = 0; Ec2 = 135 Vdc; Rl = 1,000 ohms; Eg1 = 50 Vac; Rg1 = 55,000 ohms	0.65	II	Ic2	4	10	mAdc
---	Activity	5	Ef = 2.125 Vac	0.65	II	Δep	---	10	%
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1301	Filament current	-	Ef = 2.5 V	2.5	II	If	150	180	mA
1306	Transconductance	-		2.5	II	Sm	1,400	2,300	µmhos
1316	Amplification factor (triode)	-	Eb = 150 Vdc (anode and screen tied together)	6.5	Code F	Mu	2.7	4.7	---
1230	Power oscillation	6	Class C amplifier; F = 100 MHz; Eb = Ec2 = 90 Vdc; Rg1 = 45,000 ohms	6.5	Code F	Po	0.5	---	W
1266	Primary screen grid emission	7	Eb = 0; Ec2 = 127 Vac; Ec1/Pg2 = 1 W; t = 300 sec	6.5	Code F	Is	---	200	µAdc
1331	Direct-interelectrode capacitance	-	Shield No. 316	6.5	Code F	Cg1p Cin Cout	--- 3.5 6.0	0.16 5.5 9.0	pF pF pF
1031	Low-frequency (25 Hz) vibration	-	Ef = 2.5 Vac; Ec1 = -30 Vdc; Rp = 2,000 ohms	6.5	Code F	Ep	---	100	mVac

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued								
1041	Shock	8	720 G; Ef = 2.5 Vac; Eb = 150 Vdc; Ec2 = 135 Vdc; Ec1 = -30 Vdc	6.5	---	---	---	---	---
1031	Sweep-frequency vibration	8	No voltages	6.5	---	---	---	---	---
---	Post-shock and sweep-frequency vibration test end points:			---	---	---	---	---	---
---	Inoperatives	-		---	---	If	150	180	mAdc
1301	Filament current	-		---	---	Ib	13	26	mAdc
1256	Electrode current (anode)	-		---	---	Ic2	---	2.0	mAdc
1256	Electrode current (screen)	-		---	---	Ic1	---	-1.5	µAdc
1266	Total grid current	-		---	---	Ep	---	100	mVac
1031	Low-frequency (25 Hz) vibration	-		---	---	---	---	---	---
1121	Base strain	-		2.5	Code H	---	---	---	---
(A) 2126	Glass strain	-		4.0	S3	---	---	---	---
(A) 1211	Insulation of electrodes	-		---	---	---	---	---	---
(A) 1105	Permanence of marking	-							

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL PERCENT DEFECTIVE	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3										
1501	Intermittent life	9	Group A; Ebb = 150 Vdc; Ec2 = 90 Vdc; Rl = 1,100 ohms; Ef = 1.25 Vac; Ec1 = -30 Vdc; Eg1 = 43 Vac	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours):	-									
---	Activity	-		---	---	---	---	Δep	---	15	%
---	Operation screen grid current	-		---	---	---	---	Ic2	---	15	mAdc
---	Operation peak output voltage	-		---	---	---	---	ep	85	---	V
---	Total defectives	-		---	---	1	3	---	---	---	---

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NOTES:

1. The filament shall be Wilbur-Driver Alloy No. 38, or equal, and shall have the following nominal composition: 0.8 percent aluminum, 20.5 percent tungsten, and the balance nickel.
2. For parallel filament connection, pins No. 2 and No. 6 are negative, and pins No. 4 and No. 5 are positive.
3. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
4. This test to be performed at the conclusion of the holding period.
5. Reduce filament potential of operation test and read percentage decrease of operation peak output voltage.
6. Use separately excited oscillator. Hold excitation at 35 volts peak. Load for maximum Po at Ib = 15 mAdc. Limit is for useful power output in load.
7. Pg2 = 2.48 times the product of the rectified screen current and rectified screen voltage.
8. (A) This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
9. Life test at power line frequency (see figure 2). The filament voltage and grid driving voltage may be 90 degrees out of phase. The filament and grid driving voltage should be interrupted simultaneously. The life cycling schedule shall be 1-3/4 hours "ON" and 1/4 hour "OFF."

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2618)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1358A ELECTRON TUBE, RECEIVING TYPE 3B4WA

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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